INCH-POUND

MIL-R-6106/27F
30 November 1994
SUPERSEDING
MIL-R-6106/27E
12 July 1993

MILITARY SPECIFICATION SHEET

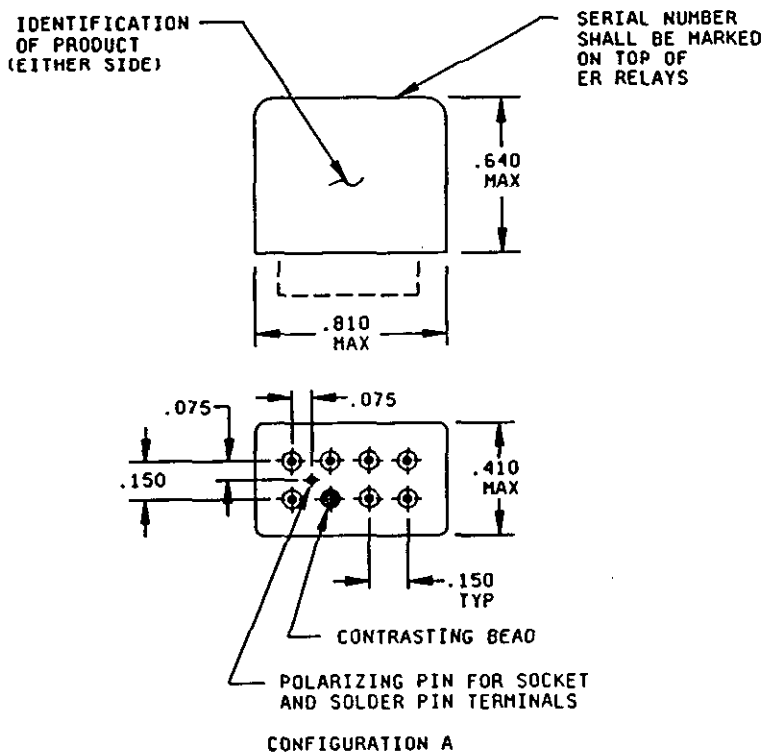
RELAYS, ELECTROMAGNETIC (INCLUDING ESTABLISHED RELIABILITY (ER) TYPES) TYPE I, 2PDT, LOW LEVEL TO 5 AMPERES, PERMANENT MAGNET DRIVE, HERMETICALLY SEALED

(F)

INACTIVE FOR NEW DESIGN AS OF 30 NOVEMBER 1994. THIS SPECIFICATION AND ITS ASSOCIATED QUALIFIED PARTS LIST ARE SCHEDULED FOR CANCELLATION IN OCTOBER 1995. REFER TO TABLE V AND MIL-R-83536 FOR SUPERSESSION DATA.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-6106.



CONFIGURATION A

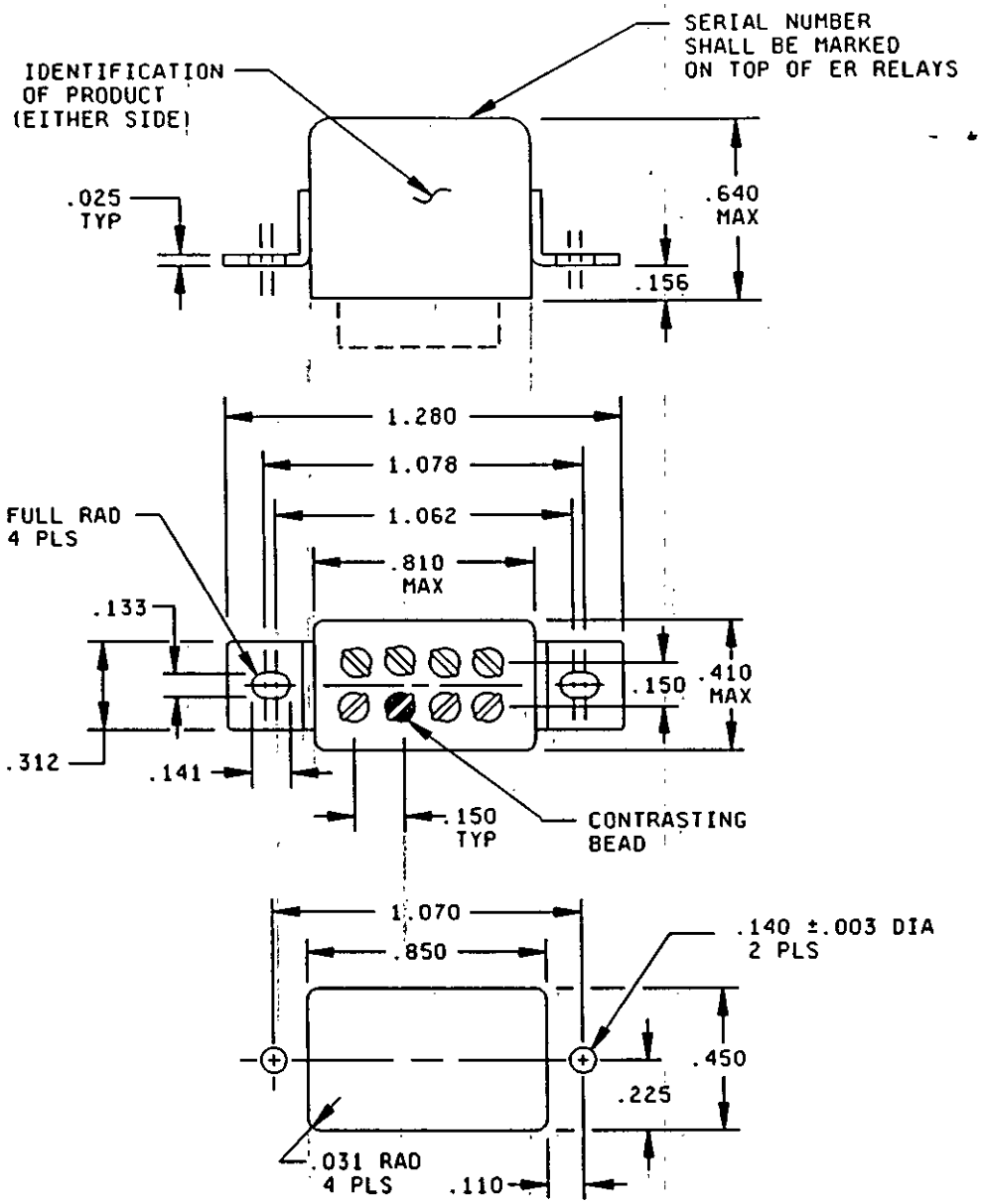
FIGURE 1. Dimensions and configurations.

(F) denotes changes

AMSC N/A FSC 5945

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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MOUNTING HOLE LAYOUT

CONFIGURATION B

FIGURE 1. Dimensions and configurations - Continued.

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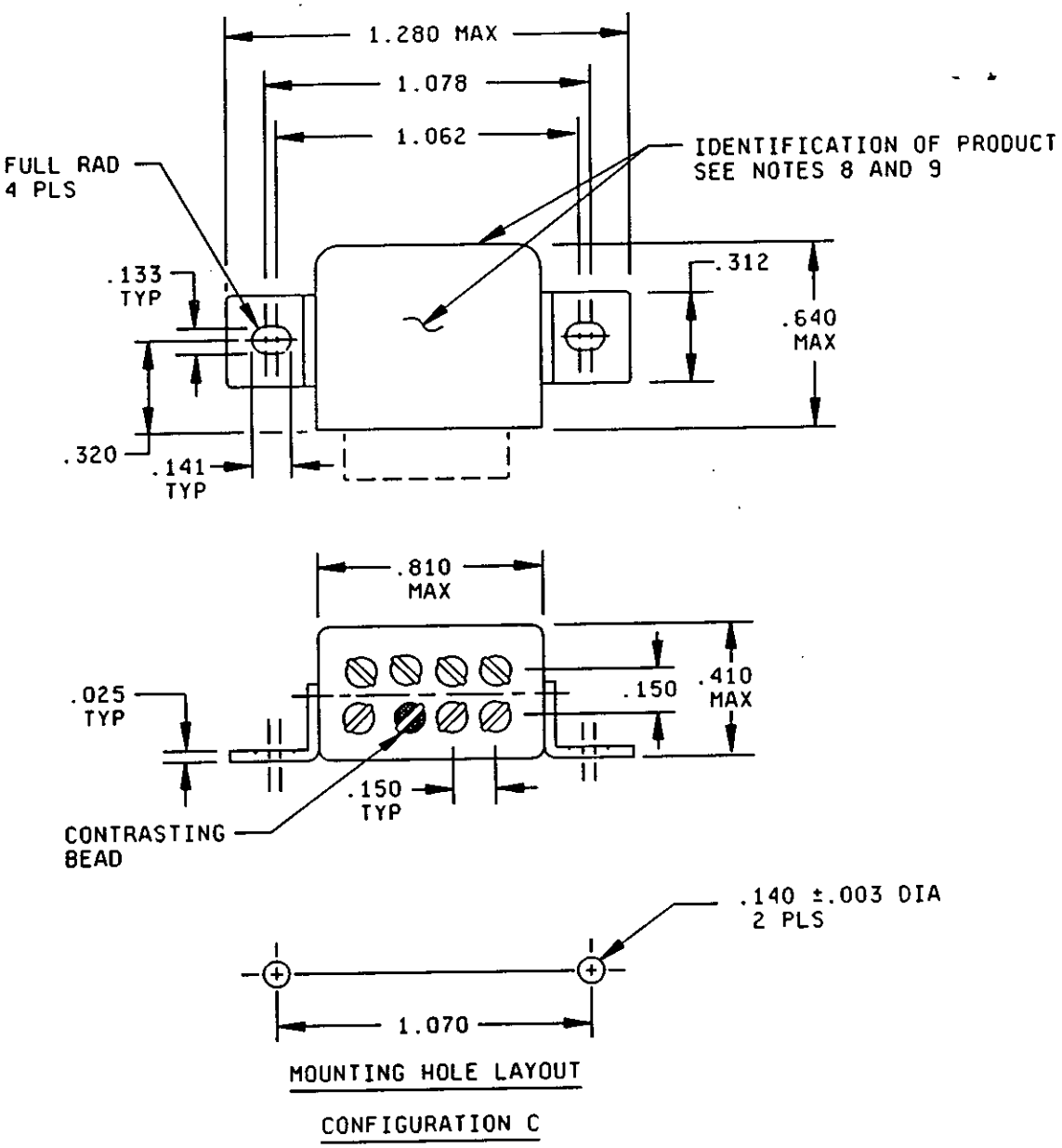


FIGURE 1. Dimensions and configurations - Continued.

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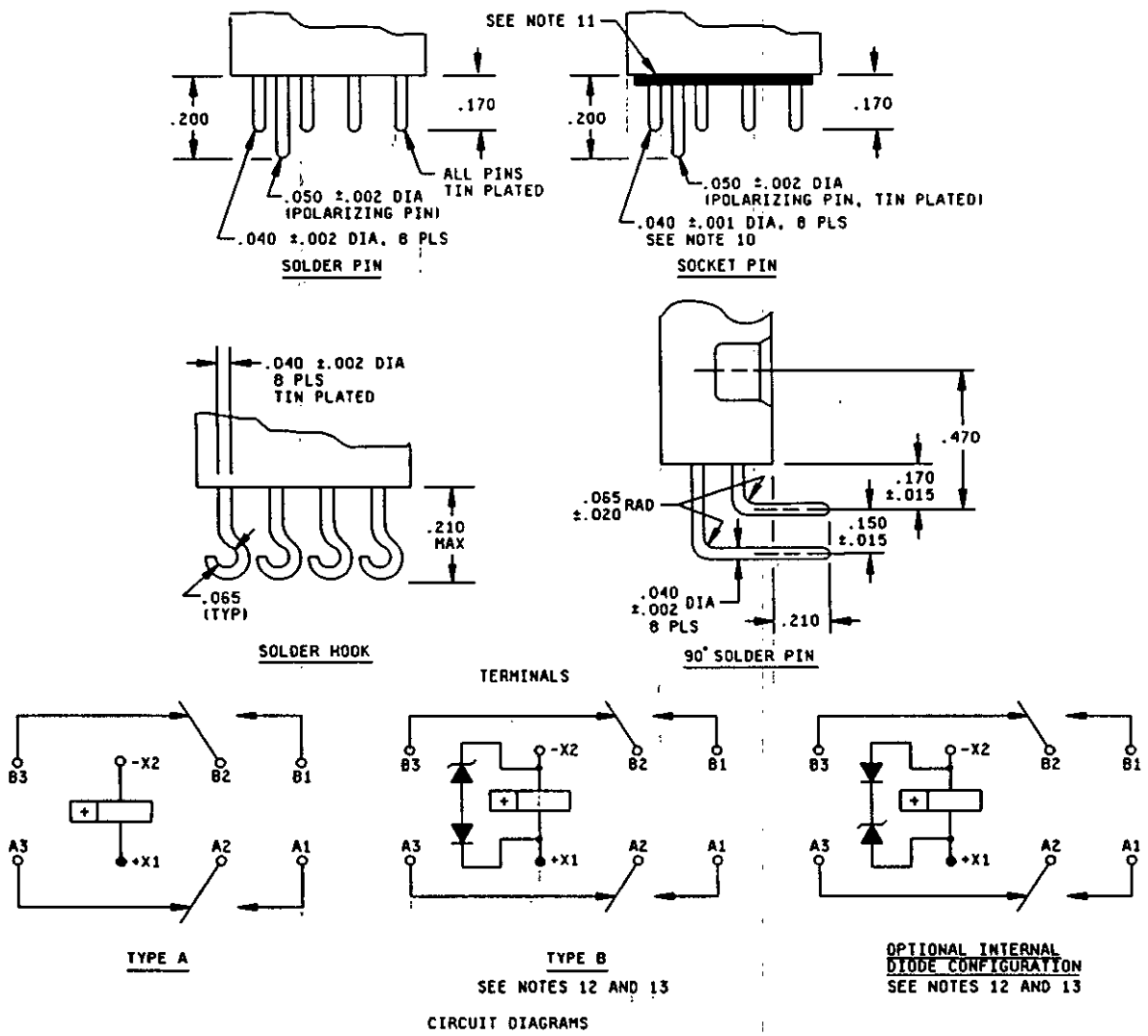


FIGURE 1. Dimensions and configurations - Continued.

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Inches	mm	Inches	mm	Inches	mm
.001	0.03	.110	2.79	.320	8.13
.002	0.05	.133	3.38	.410	10.41
.003	0.08	.140	3.56	.450	11.43
.015	0.38	.141	3.58	.470	11.94
.020	0.51	.150	3.81	.640	16.26
.025	0.64	.156	3.96	.810	20.57
.031	0.79	.170	4.32	.850	21.59
.040	1.02	.200	5.08	1.062	26.97
.050	1.27	.210	5.33	1.070	27.18
.065	1.65	.225	5.72	1.078	27.38
.075	1.90	.312	7.92	1.280	32.51

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 inch (0.25 mm).
4. There shall be affixed to the relay a legible circuit diagram that identifies each terminal location specified. These relays are polarized monostable.
5. These relays are polarized monostable.
6. This relay shall not operate or be damaged by reverse polarity. Semiconductors shall not be used for this purpose.
7. Permanent magnet drive consists of a permanent magnet with its flux path switched and combined with the electromagnetic flux.
8. Applicable to configuration C only. The circuit diagram, manufacturer's PIN, and the military PIN shall be marked on the near side. The remaining portion of the nameplate data shall be marked on the far side.
9. Applicable to configuration C only. Relays shall be marked with the manufacturer's name or source code and date code. In addition, ER relays shall be marked with the serial number. Marking shall be with the bottom of the print adjacent to the near side.
10. Socket pin terminals, except the polarizing pin, shall be gold-plated in accordance with MIL-G-45204, type II, class I. The nickel underplating shall be 50 to 150 microinches thick minimum.
11. Silicone rubber gasket, AMS 3332, shore hardness 25 ±5, thickness .050 ±.005 inch (1.27 ±0.13 mm) .
12. Back EMF (transient voltage): 42 V dc maximum.
13. JANTX or equivalent screened semiconductors shall be used in ER relays with internal coil suppression. Relays using suppression devices shall continue to operate should the suppression circuit be in a failure mode. Diodes shall have a peak inverse voltage of 600 V dc minimum.

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS:

CONTACT DATA:

Life: Unless otherwise specified: 100,000 cycles.

Load ratings:

High level (relay case grounded):

Resistive: 5 amperes at 28 V dc, 115 V ac, 400 Hz, 1 phase; 115/200 V ac, 400 Hz, 3 phase.

Inductive: 3 amperes at 28 V dc; 5 amperes at 115 V ac, 400 Hz, 1 phase and 115/200 V ac, 400 Hz, 3 phase. (Life: 20,000 cycles.)

Motor: 2 amperes at 28 V dc; 3 amperes at 115 V ac, 400 Hz, 1 phase and 115/200 V ac, 400 Hz, 3 phase.

Lamp: 1 ampere at 28 V dc.

Low level: 10 to 50 μA at 10 to 50 mV dc or peak ac. 1/ 2/ 3/

Intermediate current: Applicable.

Transfer load: Not applicable.

Mechanical life (reduced current): 1.25 amperes at 28 V dc; 115 V ac, 1 phase, 400 Hz; and 115/200 V ac, 3 phase, 400 Hz. (Life: 400,000 cycles.)

Mixed loads: Applicable. 4/

Contact voltage drop or resistance: 1/ 2/ 3/

High level:
Initial: 0.150 volt maximum.
After life: 0.175 volt maximum.

Low level:
Initial: 0.050 ohm maximum.
After life: 0.150 ohm maximum.

Contact bounce: 0.5 millisecond maximum.

Break bounce (normally open contacts only): 0.1 millisecond maximum (applicable to dc suppressed coils only).

1/ For low level, the following shall apply:
a. During endurance and operational reliability testing, contact load shall be 10 to 50 microamperes; 10 to 50 millivolts open circuit voltage, 100 ohms maximum contact resistance.
b. Static contact resistance shall be performed at 50 mA maximum, 50 mV maximum.

2/ For all ER relays, the alternate low level test of operational reliability shall be used.

3/ For the group A contact voltage drop test, high level testing shall be performed first, followed by low level testing. The contacts shall not make or break the high level load.

4/ Relay shall be capable of switching low level while switching any of the other rated loads on adjacent contacts.

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Overload current: 20 amperes dc.
30 amperes ac.

Rupture current: 25 amperes dc.
40 amperes ac.

Time current relay characteristics: Applicable (see table I).

TABLE I. Time current relay characteristics. 1/ 2/

1	10 A	1 hour
2	15 A	20 seconds
3	37.5 A	2 seconds
4	75 A	.53 second

1/ CAUTION: Compare with time current characteristics of the associated circuit protective device.

2/ All relays shall withstand overload and fault currents. Relays must be able to sustain five applications (make and carry only) of power concurrently on adjacent poles at each of four different current levels in the sequence listed in table I. Separate relays shall be tested at 28 V dc and 115/200 V ac, 400 Hz, 3 phase. Cooling time between successive applications shall be 30 minutes. Tests shall be performed on both normally open and normally closed contacts. There shall be no failures or evidence of welding or sticking and relays shall pass contact voltage drop at the conclusion.

COIL DATA 5/: (See table II).

Operate time: 4 milliseconds maximum with rated coil voltage.

Release time: 4 milliseconds maximum from rated coil voltage.

Duty rating: Continuous.

ELECTRICAL DATA:

Insulation resistance, initial: 100 megohms.

After life or environmental tests: 50 megohms.

Dielectric withstanding voltage (sea level):	Initial V rms (60 Hz)	After life tests V rms (60 Hz)
Coil to case	1,000	750
All other points	1,000	750

Dielectric withstanding voltage (altitude) 6/:	80,000 ft V rms (60 Hz)	300,000 ft V rms (60 Hz)
Coil to case	250	500
All other points	250	500

5/ CAUTION: Due to possible interaction of relay magnetic fields, the following spacing requirements, as a minimum, shall be considered in dense packaging situations:
a. Row to row assisting fields: .125 inch.
b. Row to row opposing fields: .1875 inch.
c. Side to side alternating fields: .0625 inch.
d. Side to side like fields: .125 inch.

6/ Dielectric may be improved by suitable insulation of terminals and wiring after installation.

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TABLE II. Dash numbers and characteristics. 1/

PIN M6106/27- 2/ 3/	Type	Terminal type	Mounting configuration	Circuit type	Coil data								
					Rated voltage V dc 4/	Resistance Ω ±10% at 25°C	Maximum		Pickup voltage (max) (V dc)			Hold voltage V dc 6/	Drop-out voltage V dc 6/
							Voltage V dc 5/	Current A	Normal 6/	High temp test	Continuing current test		
001 7/	I	Solder pin	A	A	28	500	29	.065	18	19.8	22.5	7.0	1.5
002	I ER	"	"	A	"	"	"	"	"	"	"	"	"
003 7/	I	"	"	B	"	"	"	"	"	"	"	"	"
004	I ER	"	"	B	"	"	"	"	"	"	"	"	"
005 7/	I	Solder hook	"	A	"	"	"	"	"	"	"	"	"
006	I ER	"	"	A	"	"	"	"	"	"	"	"	"
007 7/	I	"	"	B	"	"	"	"	"	"	"	"	"
008	I ER	"	"	B	"	"	"	"	"	"	"	"	"
009 7/	I	Socket pin	"	A	"	"	"	"	"	"	"	"	"
010	I ER	"	"	A	"	"	"	"	"	"	"	"	"
011 7/	I	"	"	B	"	"	"	"	"	"	"	"	"
012	I ER	"	"	B	"	"	"	"	"	"	"	"	"
013 7/	I	Solder pin	B	A	"	"	"	"	"	"	"	"	"
014	I ER	"	"	A	"	"	"	"	"	"	"	"	"
015 7/	I	"	"	B	"	"	"	"	"	"	"	"	"
016	I ER	"	"	B	"	"	"	"	"	"	"	"	"
017 7/	I	Solder hook	"	A	"	"	"	"	"	"	"	"	"
018	I ER	"	"	A	"	"	"	"	"	"	"	"	"
019 7/	I	"	"	B	"	"	"	"	"	"	"	"	"
020	I ER	"	"	B	"	"	"	"	"	"	"	"	"
021 7/	I	Socket pin	"	A	"	"	"	"	"	"	"	"	"
022	I ER	"	"	A	"	"	"	"	"	"	"	"	"
023 7/	I	"	"	B	"	"	"	"	"	"	"	"	"
024	I ER	"	"	B	"	"	"	"	"	"	"	"	"
025 7/	I	Solder pin	C	A	"	"	"	"	"	"	"	"	"
026	I ER	"	"	A	"	"	"	"	"	"	"	"	"
027 7/	I	"	"	B	"	"	"	"	"	"	"	"	"
028	I ER	"	"	B	"	"	"	"	"	"	"	"	"
029 7/	I	Solder hook	"	A	"	"	"	"	"	"	"	"	"
030	I ER	"	"	A	"	"	"	"	"	"	"	"	"
031 7/	I	"	"	B	"	"	"	"	"	"	"	"	"
032	I ER	"	"	B	"	"	"	"	"	"	"	"	"
037 7/	I	90° solder pin	"	A	"	"	"	"	"	"	"	"	"
038	I ER	"	"	A	"	"	"	"	"	"	"	"	"
039 7/	I	"	"	B	"	"	"	"	"	"	"	"	"
040	I ER	"	"	B	"	"	"	"	"	"	"	"	"

1/ Each relay possesses both high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits are not recommended for subsequent use in low level applications.

2/ For type I ER relays, a suffix letter (M, U, X, or P), to indicate the applicable failure rate level, shall be added to the applicable dash number. Failure rate level (percent per 10,000 cycles): M = 1.0 percent, U = 0.5 percent, X = 0.3 percent, and P = 0.1 percent. Example: M6106/27-002M; M6106/27-040P.

3/ M6106/27-033 through M6106/27-036 are canceled without replacement.

4/ CAUTION: Use of any coil voltage less than rated coil voltage will compromise the operation of the relay.

5/ When maximum ambient temperature does not exceed +85°C, the maximum coil voltage shall be 32 V dc.

6/ Over the temperature range.

7/ Type I relays are inactive for new design after 14 December 1990. See table IV for cross-reference data.

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ENVIRONMENTAL CHARACTERISTICS:

Temperature range: -70°C to +125°C.

Maximum altitude rating: 300,000 feet.

Shock (specified pulse): MIL-STD-202, method 213, test condition C, except peak value shall be 200 g's. Contact chatter shall not exceed 10 microseconds for closed contacts and 1 microsecond maximum for open contacts.

Vibration (sinusoidal): Applicable; 30 g's maximum and frequency range shall be 70 to 3,000 Hz. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum for open contacts.

Vibration (random): MIL-STD-202, method 214, test condition IG, 15 minutes each plane.

Acceleration: Applicable (15 g's maximum).

PHYSICAL DATA:

Dimensions and configurations: See figure 1.

Weight (maximum): .034 pound (15.4 grams).

Terminal strength:

Solder hook terminals:

Twist test: Not applicable.

Pull force: 5 ±0.5 pounds.

Solder pin terminals:

Bend test: Applicable to 0.040 diameter pins.

Pull force (0.040 diameter): 5 ±0.5 pounds.

Pull force (0.050 diameter): 10 ±1 pounds (applicable to polarizing pin only).

Plug-in terminals:

Bend test: Applicable to 0.040 diameter pins.

Pull force (0.040 diameter): 5 ±0.5 pounds.

Pull force (0.050 diameter): 10 ±1 pounds (applicable to polarizing pin only).

Terminal solderability: Applicable to solder pin and solder hook terminals only.

Seal: Hermetic; relays shall be sealed by welding (laser, TIG (tungsten inert gas) or other suitable means as approved by the qualifying activity).

Construction (internal and external): All welded, except that coil magnet wire to coil lead wire shall be soldered.

Identification of product: Applicable.

Quality conformance inspection: Performance of group C inspection for Type I and Type I-ER relays is not required. Performance of Retention of Qualification for Type I-ER relays is not required, provided the manufacturer possesses and maintains a current qualification to MIL-R-83536.

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Qualification by similarity: See table III.

TABLE III. Qualification by similarity. 1/

PIN M6106/27-	Loads		Dynamics			Environmental
	Basic qualification	ER	Sockets/pins			
	A	D	A	B	C	A
	2/ 3/ DC	2/ 3/ DC	2/ 3/ Configuration A	2/ 3/ Configuration B	2/ 3/ Configuration C	2/ 4/ DC
001	4	3	2			4
002	4	3	2			4
003	1, 2, 3	1, 2	1			1, 2, 3
004	2, 3	1, 2	1			2, 3
005	4	3	2			4
006	4	3	2			4
007	1, 2, 3	1, 2	1			1, 2, 3
008	2, 3	1, 2	1			2, 3
009	4	3	2			4
010	4	3	2			4
011	1, 2, 3	1, 2	1			1, 2, 3
012	2, 3	1, 2	1			2, 3
013	4	3		2		4
014	4	3		2		4
015	1, 2, 3	1, 2		1		1, 2, 3
016	2, 3	1, 2		1		2, 3
017	4	3		2		4
018	4	3		2		4
019	1, 2, 3	1, 2		1		1, 2, 3
020	2, 3	1, 2		1		2, 3
021	4	3		2		4
022	4	3		2		4
023	1, 2, 3	1, 2		1		1, 2, 3
024	2, 3	1, 2		1		2, 3
025	4	3			2	4
026	4	3			2	4
027	1, 2, 3	1, 2			1	1, 2, 3
028	2, 3	1, 2			1	2, 3
029	4	3			2	4
030	4	3			2	4
031	1, 2, 3	1, 2			1	1, 2, 3
032	2, 3	1, 2			1	2, 3
037	4	3			2	4
038	4	3			2	4
039	1, 2, 3	1, 2			1	1, 2, 3
040	2, 3	1, 2			1	2, 3

1/ See amendment 1, MIL-R-6106, for guidelines on determining how relays are to be grouped and ranked within each subgroup.
2/ DC suppressed coils: Discretes qualify only discretes; hybrids qualify only hybrids.
3/ Terminals: Socket pin type terminals are considered worst case. Solder pins and hooks may be qualified by similarity.
4/ Socket pins with gasket seals are considered nonsimilar to solder pins and hooks, and each must be tested separately.

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(F) Cross-reference for Government logistics support: See table IV.

(F) TABLE IV. Cross-reference for Government logistics support.

PIN M6106/27-	Support with PIN M83536	PIN M6106/27-	Support with PIN M83536
001	/1-019	021	/1-024
002	/1-019	022	/1-024
003	/2-019	023	/2-024
004	/2-019	024	/2-024
005	/1-020	025	/1-025
006	/1-020	026	/1-025
007	/2-020	027	/2-025
008	/2-020	028	/2-025
009	/1-021	029	/1-026
010	/1-021	030	/1-026
011	/2-021	031	/2-026
012	/2-021	032	/2-026
013	/1-022	033	1/
014	/1-022	034	1/
015	/2-022	035	1/
016	/2-022	036	1/
017	/1-023	037	/1-027
018	/1-023	038	/1-027
019	/2-023	039	/2-027
020	/2-023	040	/2-027

1/ M6106/27-033 through -036 are canceled without replacement.

(F) Supersession data: See table V.

(F) TABLE V. Supersession data.

Superseded PIN M6106/27-	Replacement PIN M83536	Superseded PIN M6106/27-	Replacement PIN M83536
001	/1-019	021	/1-024
002	/1-019	022	/1-024
003	/2-019	023	/2-024
004	/2-019	024	/2-024
005	/1-020	025	/1-025
006	/1-020	026	/1-025
007	/2-020	027	/2-025
008	/2-020	028	/2-025
009	/1-021	029	/1-026
010	/1-021	030	/1-026
011	/2-021	031	/2-026
012	/2-021	032	/2-026
013	/1-022	033	1/
014	/1-022	034	1/
015	/2-022	035	1/
016	/2-022	036	1/
017	/1-023	037	/1-027
018	/1-023	038	/1-027
019	/2-023	039	/2-027
020	/2-023	040	/2-027

1/ M6106/27-033 through -036 are canceled without replacement.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-0940-01)

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